

Effects of Street Design on Pedestrian Safety

Table of Contents

Introduction	3
Literature Review	3
Excessive Vehicle Speed	3
Traffic Calming Measures	5
Crosswalk Implementation	6
Research Methodology	8
Hypothesis	8
Overview	8
Analysis	8
Results	10
Implications	12
Limitations and Improvements	12
Conclusion	14
Bibliography	16

Introduction

In 2022, 7,522 pedestrians were killed in collisions with motor vehicles in the United States alone, and another 67,000 pedestrians were injured. 1,705 fatalities occurred at intersections. While it may appear that infrastructure for pedestrians is always improving, fatalities have risen significantly in the past fifteen years from their all-time low of 4,109 (IIHS). Pedestrians are one of the most vulnerable groups of road users because they lack the physical protections that those in automobiles benefit from. As such, collisions with pedestrians are far more likely to result in severe injury or death. Because the number of pedestrian fatalities is still increasing, it is imperative that street designers find ways to better protect pedestrians to achieve the “Vision Zero” goals that many countries are now implementing (Shahum). Many different street design elements exist to protect pedestrians including curb extensions; marked and signalized crosswalks; speed reduction measures such as speed bumps; and others. These serve two primary purposes: “calming” motor vehicle traffic (this primarily aims to reduce speed) and increasing pedestrian visibility. Both of these measures make drivers more aware of the presence of pedestrians. Crosswalks are one of the most common of these design elements, and they can be found in many areas where pedestrian traffic and vehicle traffic are likely to coincide. Unfortunately, crosswalk coverage in the United States is still quite limited; even in San Francisco, a relatively dense and walkable city, crosswalk coverage is only 58% (Moran). This issue is even worse in less populated cities, especially those in the US Midwest where improvements in motor vehicle infrastructure are prioritized over pedestrian safety measures. This paper aims to determine how street design elements can affect the probability and severity of pedestrian and motor vehicle collisions as well as the extent to which these elements are already present in streets in the US Midwest.

Literature Review

Excessive Vehicle Speed

Pedestrian injuries and fatalities are a problem that is only becoming worse. According to an article from the Insurance Institute for Highway Safety, “Deaths from motor vehicle crashes with pedestrians have increased 83% since reaching their low point in 2009 and account for 18% of crash fatalities” (“Fatality Facts”). This situates the importance of improving pedestrian facilities because it demonstrates that current progress is not sufficiently alleviating the problem of pedestrian injuries and fatalities. Another article from the Insurance Institute of Highway

Safety, “Pedestrians and Bicyclists,” provides an overview of effective methods. The article states that reducing vehicle speed, adding elements to the street such as crosswalks and curb extensions, giving pedestrians sufficient time at traffic signals, and altering vehicle design to reduce crash severity are all ways to reduce the likelihood and severity of pedestrian-vehicle collisions (“Pedestrians and Bicyclists”).

Vehicle speed has been repeatedly shown to be a major factor in both the likelihood and severity of collisions. According to Brian Tefft, a Senior Research Associate at the AAA Foundation for Traffic Safety, “Results show that the average risk of severe injury for a pedestrian struck by a vehicle reaches 10% at an impact speed of 16 mph, 25% at 23 mph, 50% at 31 mph, 75% at 39 mph, and 90% at 46 mph. The average risk of death for a pedestrian reaches 10% at an impact speed of 23 mph, 25% at 32 mph, 50% at 42 mph, 75% at 50 mph, and 90% at 58 mph” (Tefft 1). Higher vehicle speeds result in more severe collisions, and many other studies found very similar results through analysis of accident data recorded by governments, insurance agencies, and researchers themselves. This study demonstrates that speed has a substantial impact on the possibility for severe injury or death.

An article from the Institute for Road Safety Research (SWOV) in the Netherlands, “The Relationship Between Speed and Crashes,” found that “at a collision speed of 20 km/h nearly all pedestrians survive a crash with a passenger car; about 90% survive at a collision speed of 40 km/h, at a collision speed of 80 km/h the number of survivors is less than 50%, and at a collision speed of 100 km/h only 10% of the pedestrians survive.” These results are very close to the findings of Tefft and others, showing that higher speeds greatly increase the risk of pedestrian collisions, injuries, and fatalities (“Speed and Crashes”). Another article from the SWOV found that greater vehicle mass contributed to pedestrian accident severity, which is particularly a problem in North America due to the popularity of larger and heavier vehicles. The source found that speeding was a major factor in the likelihood of collisions, stating that “about one third of fatal crashes are partly due to speeding or to inappropriate speeds” (“Speed Management” 7). Due to the limitations of regulations in actually changing driver behavior, it is important for road designers to use physical features of streets to encourage safe speeds and safe driver behavior. Almost identical results were found by D.C. Richards, a researcher at the London Department for Transit who analyzed crash severity based on speed (Richards).

Traffic Calming Measures

Leah Shahum found similar limitations due to driver behavior in her paper “Safe Streets: Insights on Vision Zero Policies from European Cities.” Shahum emphasizes that road designers cannot assume that road users will comply with signs and regulations, to design features of the roadway must be used “to encourage an appropriate — and inherently safe — level of speed for users” (Shahum). The features described in Shahum’s paper aligned with those found by others; she states that “the width of travel lanes; . . . radius of turns; . . . raised crosswalks and corner bulb-outs, . . . and the use of well-designed traffic circles over traffic stop lights” (Shahum) are all ways to calm traffic, thereby bettering conditions for pedestrians.

The roadway design elements described by Shahum are almost identical to those Herman F. Huang and Michael J. Cynecki found effective in their paper “Effects of Traffic Calming Measures on Pedestrian and Motorist Behavior.” Huang and Cynecki state that the aforementioned pedestrian facilities “may slow vehicle speeds, thus increasing the number of adequate gaps and also increasing the likelihood that an approaching motorist will yield to a pedestrian” (Huang and Cynecki 28). Taking both sources as a whole, it is clear that the measures the researchers encourage would be highly beneficial in improving driver yielding and pedestrian safety.

In addition to high vehicle speed, the time that a pedestrian spends in a crosswalk also affects injury severity, according to Sylvia Zajac and John Ivan. The researchers found that “clear roadway width (the distance across the road including lane widths and shoulders, but excluding the area occupied by on-street parking)” was one significant factor in pedestrian injuries and fatalities, among others (Zajac and Ivan). While the accident severity was, in this case, linked to the time pedestrians spent crossing the intersection, roadway width also affects injury severity because of the effect it has on vehicle turning speed.

According to Kendra K. Levine, a researcher at the Institute of Transportation Studies Library of the California Department of Transportation, “In a recent paper, [the National Complete Streets Coalition] recommend reducing traffic speed by tightening curb radii to the minimum feasible design vehicle. Possible encroachment or lane departure would not be as dangerous due to the reduced speeds.” This means that driver error and visibility issues would be less likely to result in severe collisions with other motor vehicles and pedestrians, thereby increasing pedestrian safety (Levine 4).

The Oregon Department of Transport also recommends reducing curb radius to lower vehicle turning speeds in the street design manual *Main Street: When A Highway Runs Through It*. The source states that “On most older traditional main streets, the corners are often very sharp, with a small corner radius; this is one of the most effective tools for controlling speed. A vehicle turning a corner has to clear the curb, and the sharper the corner, the slower the vehicle must go” (*Main Street* 41). In addition to lowering turning speeds, it is also imperative that pedestrians are visible to drivers so that drivers can anticipate pedestrian crossings. The *Main Street* handbook accounts for this by stating that “two parallel painted lines are generally not enough of a distinguishing marking for crosswalks” (*Main Street* 42).

Randall S. Johnson, a researcher at the Oregon Department of Transportation, found that curb extensions are an effective method for both reducing crossing distance and calming traffic. Johnson studied one intersection before and after the addition of a curb extension. By determining how many drivers yielded to pedestrians before and after the installation of the curb extension, Johnson found that “The analysis of the average number of vehicles that pass before a pedestrian-cross showed fewer passing for crossings from the side with the curb extension compared to crossings from the unimproved side” (Johnson 13). This study shows that the addition of simple pedestrian facilities such as curb extensions can have a significant impact on driver behavior.

Although curb extensions can be highly beneficial, as with other pedestrian facilities, they must be implemented properly to have a significant effect. According to a meta-analysis by Niel Florian of curb extension regulations in various countries in Europe, “Curb extensions should at least reach out to the lane. Smaller extensions than the parking lane should not be built. There is no maximum width in relation to the parking lane.” Thus, increasing curb extension width will always be beneficial, as it further reduces vehicle speed (Florian).

Crosswalk Implementation

While crosswalk design can increase pedestrian safety in a well-lit environment, markings can become virtually invisible at night without sufficient lighting. As a result, the number of pedestrian injuries and fatalities that occur at night is disproportionately high compared to daytime conditions. According to Rajaram Bhagavathula and Ronald Gibbons, seventy-five percent of pedestrian fatalities in the United States, despite having only twenty-five percent of daytime traffic volumes. The researchers state that “Since approximately 90% of the

information drivers use to navigate the roads is visual, avoiding pedestrians becomes more challenging with less light.” The researchers therefore suggested that road designers use lighted crosswalks and bollard-level lighting--among other methods--to greatly improve pedestrian visibility and the ability of drivers to respond to pedestrians crossing at crosswalks (Bhagavathula and Gibbons).

Brian J. Crowley-Koch and Ron Van Houten studied the effects of signals by pedestrians on vehicles yielding. While pedestrian signals are not a physical element of the streetscape, the study shows that increasing pedestrian visibility can be beneficial for prompting motorists to yield. The study also found that yielding was much lower when speeds were higher (Crowley-Koch and Van Houten). Although the aim of the study was not to determine how speed affects yielding, these results are important to developing an understanding of how reduced speeds can have an impact on pedestrian safety.

As with poor visibility, pedestrian facilities cannot be effective if they do not exist. According to M.E. Moran, there are usually more crosswalks in high-income areas and signalized intersections. Thus, the lack of crosswalks disproportionately affects lower-income and marginalized communities. Additionally, road designers often install crosswalks in response to high pedestrian volumes instead of installing them to encourage pedestrian traffic and improve pedestrian safety. Moran suggests that “planners should move from installing crosswalks ‘reactively’ (or only where warrants are met), and transition to ‘proactive’ crosswalk installation in order to spur walking” (Moran).

A study from Patrick Bowen in 2016 was used while analyzing intersections in Evansville. This study provides details about demographic data that is separated by census tract and covers the entire city of Evansville. It contains details about poverty rates, home occupancy, crime index, tax delinquency, and many other factors to create a “final score” indicating the condition of an area (Bowen). This data was useful because it facilitated a comparison of crosswalk coverage to factors such as the poverty rate of a census tract.

Research Methodology

Hypothesis

The researcher speculated that low-income areas would have lower crosswalk coverage and poorer crosswalk conditions based on the finding of other researchers such as M.E. Moran. Additionally, it was speculated that a greater distance from the city center would decrease crosswalk coverage because these areas would have a lesser concentration of businesses and a lower residential and commercial density.

Overview

Congruent with the methods used in several sources for this paper, a case study of 3,536 intersections using both quantitative and qualitative analysis was conducted to gain insight into the distribution of intersections in Evansville, Indiana, a city in the Midwest with a population of approximately 140,000. To conduct this study, intersections distributed across 38 census tracts in Evansville were individually analyzed. Each intersection was evaluated for its crosswalk “completeness” and condition, two factors that greatly affect the walkability and pedestrian safety of an area (Moran). The crosswalk coverage data was then compared to other factors such as the poverty rate of census tracts or their distance from the city center to determine what factors affect an area’s crosswalk coverage.

Analysis

The primary source for data in this study was Google Maps, which was used to visually examine every intersection in Evansville. This only required access to a computer with an internet connection and a spreadsheet to record data. The methodology used for analysis was similar to that of M.E. Moran in “Where the Crosswalk Ends” but it was simplified to reduce the amount of time needed for each intersection. While Moran’s study required approximately one minute of analysis per intersection, this study required only a few seconds for incomplete intersections, with the most complete intersections still taking less than fifteen seconds. Previous studies such as Moran’s included geospatial data for each intersection by recording the street names, but this study excluded these elements and only recorded which census tract the intersection was in.

Two main factors were analyzed for intersections in each census tract: “completeness,” or the fraction of crosswalks per intersection, and the overall condition of pedestrian facilities at

each intersection. For the former, two factors were recorded to determine completeness. First, the number of converging streets, or “branches” of an intersection was recorded in one column of a spreadsheet. Four-way intersections were given a “4” in this column, while T-junction intersections had a “3.” In the next column, the number of crosswalks in the corresponding intersection was recorded. This meant that the average crosswalk coverage could be found as a percentage.

In addition to completeness, the condition of each intersection was also recorded. Crosswalks were either rated 1, indicating work markings, or 2, indicating that the markings were in good condition. If markings were worn to the point of being only slightly visible, they were not counted as a crosswalk at all, so their condition was not recorded.

The results for each intersection included three numbers that represented branches, crosswalks, and crosswalk condition. Thus, a “complete” four-branch intersection might have the numbers 4-4-2, while a three-branch intersection lacking pedestrian facilities would have the numbers 3-0-0 (the zeroes would appear as blank values in the spreadsheet).

Several other considerations had to be made to ensure that intersections would be analyzed properly in spite of the lack of geospatial data in this study. For example, intersections on the edges of census tracts could be analyzed twice if they were included in the next census tract. Thus, intersections along the borders of census tracts would only be analyzed for the first tract, and they would be excluded in the next. This was possible because census tracts were analyzed sequentially, starting at 18163000**1.00** and ending at 181630**102.02** (some numbers were skipped because those tracts do not lie within the city limits). Other possibilities were considered to do this, such as analyzing the north and west branches of intersections in one tract and the south and east branches of the next. However, this was deemed too complex and error-prone compared to the simpler solution that was implemented in this study.

The distance of each census tract from the city center was also recorded so that crosswalk coverage could be represented as a function of this distance. This was done by measuring the distance from the Civic Center in downtown Evansville to the closest point of each census tract. The Civic Center was selected because it is located in the densest part of the city and is a major landmark, and the “Measure Distance” feature in Google Maps was used to collect this data.

Results

The average crosswalk coverage for the city of Evansville was 8.29% and the average condition on a scale from 1 to 2 was 1.53. This result was within the expected range given that most parts of Evansville are not designed for heavy pedestrian traffic. The highest crosswalk coverage for a single census tract was 45.29% in downtown Evansville (census tract 18.00). The far higher crosswalk coverage in this area was likely due to its pedestrian-friendly design. Census tract 32.00 had no crosswalks, which may be due to its lack of businesses, low density, and distance from the city center. Unexpectedly, census tracts 3, 4, and 14 had relatively low crosswalk coverage despite containing Evansville's Walnut Street Improvement Project, which increased crosswalk coverage along one thoroughfare but had little influence on the area's overall crosswalk coverage ("Walnut Street Improvement Project").

Overall, the most crosswalks were present in dense areas where pedestrians would be more likely to use the facilities provided. A very weak to nonexistent correlation was found between the economic conditions in an area and its crosswalk coverage, as evidenced by Fig. 1.

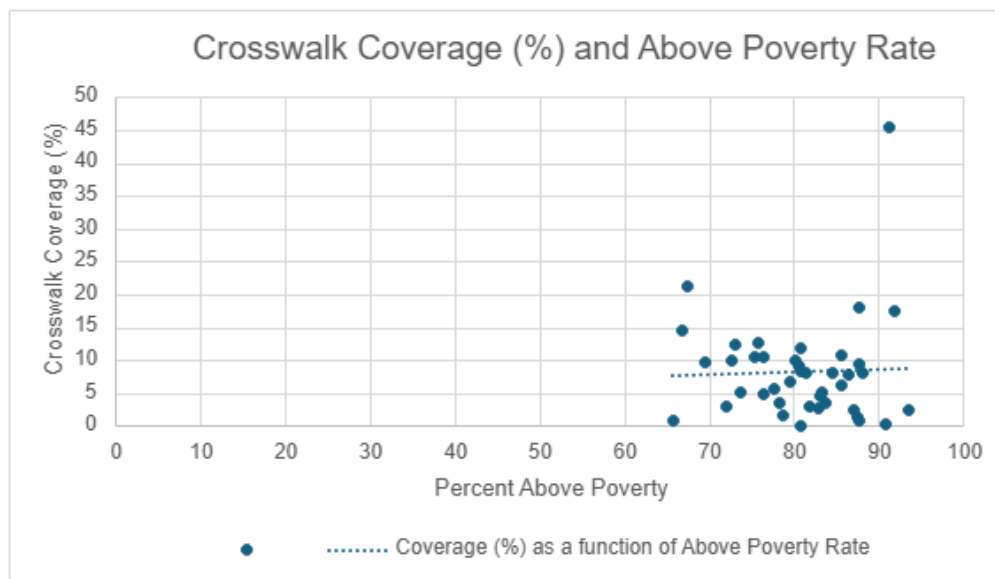


Fig. 1. Graph of Crosswalk Coverage and Above Poverty Rate

The extremely weak correlation between crosswalk coverage and poverty rate demonstrates that crosswalks are not distributed based on the socioeconomic conditions of an area. This result does not align with the original hypothesis, but several factors could explain the lack of correlation

between these variables. First, the poverty rate of an area does not necessarily determine what type of development it present in that area, as an area with a low poverty rate could also be a low density area with fewer urban crossings, while an area with a high poverty rate could be in a dense area, necessitating more crosswalks. Furthermore, cities may make attempts to spur economic growth by building more pedestrian facilities, as these would make the area more appealing for new businesses and residents.

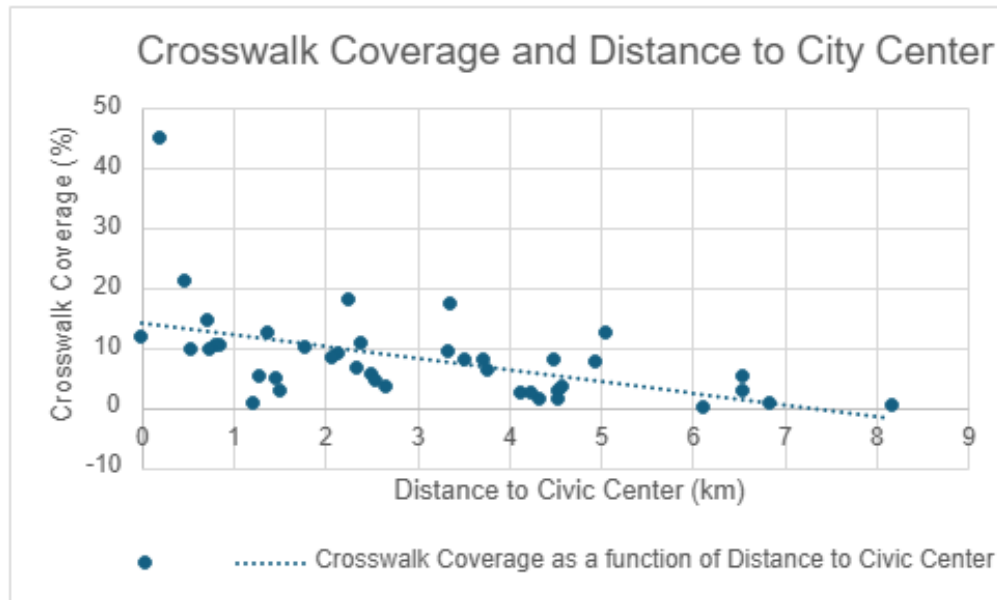


Fig. 2. Graph of Crosswalk Coverage as a Function of Distance to Civic Center

Unlike poverty rate and crosswalk coverage, it appears that the distance of a census tract from the city center affects that area's crosswalk coverage. In this case, for every additional kilometer from the city center, crosswalk coverage decreased by 1.94% on average. This result is consistent with the hypothesis that crosswalk coverage would decrease with greater distance to the city center and could be explained by the lack of businesses and other features that would compel city planners to install more crosswalks.

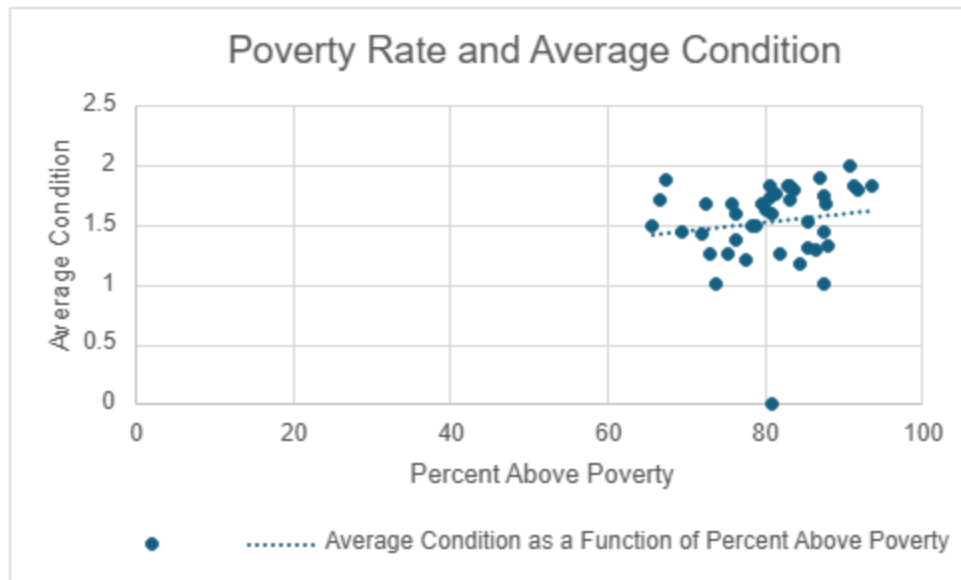


Fig. 3. Graph of Poverty Rate and Crosswalk Condition

There was a weak positive correlation between the proportion non-impovertised residents and crosswalk condition, indicating that crosswalks in higher-income areas may be of a better condition than those in low-income areas. This result was consistent with the hypothesis; however, the weakness of the correlation between the two variables should be considered when examining this data, and the results of this study may not be observed in other cities.

Implications

Although crosswalk coverage is low at 8.29%, crosswalks are relatively evenly distributed throughout the city, and it appears that the socioeconomic conditions of an area do not affect the number of crosswalks present. Combined with the literature used in this paper, the study indicates that crosswalk coverage is insufficient and that improvements in pedestrian safety could be brought about by building more crosswalks. This is particularly the case in suburban parts of Evansville, where crosswalk coverage can be as low as 0%. Even though these areas might not have as much pedestrian traffic, pedestrians could benefit from the addition of crosswalks, which are relatively inexpensive. On the contrary, the even distribution of crosswalks is a positive because it indicates that Evansville's decision to install crosswalks does not depend on the economic conditions of the area they are being installed in.

Limitations and Improvements

As stated previously, the methodology used in this study was much simpler than that of previous studies. This introduced inherent limitations and possibilities for error. For example, the

lack of geospatial data for individual intersections meant that it was possible to miss any intersection or record it twice, which warranted careful examination of each intersection. Attempts to reduce error were made by starting at the northwest corner of each census tract and going east along parallel streets; however, it is likely that some intersections were analyzed twice or skipped entirely. Furthermore, this aspect of the methodology required that census tracts be analyzed at one time to avoid errors.

Another problem with this study occurred as a result of the way intersections at the edges of census tracts were analyzed. Census tracts are usually divided along major roads, and these roads often contain many of the intersections that are a part of a census tract. Thus, the first census tracts recorded may have had a higher crosswalk coverage than later ones because they included more major roads. One possible way to alleviate this would have been to analyze half of each intersection that bordered another census tract, but this would have likely introduced the possibility for more error depending on the way intersections were divided, and such a method was deemed unsuitable for this study.

This study also did not analyze types of crosswalks, a factor that can change their effectiveness. Common crosswalk types include standard crosswalks (two parallel lines perpendicular to the street), continental crosswalks (also called zebra crossings; these are lines parallel to the street), and ladder crosswalks (a combination of standard and continental crosswalks) (Moran). Recording the type of crosswalk can provide further insight into the potential effectiveness of pedestrian safety measures.

The scope of this study was quite narrow because it only focused on one city in the US Midwest. This means that differences between cities could not be evaluated, and only areas within one city could be compared. This is a limitation because there are almost certainly variations in crosswalk coverage in different cities based on factors such as the density of the city and the decisions of local governments. There is no legal requirement to install pedestrian facilities in certain areas, and they are often installed as a response to accidents, not to prevent them. (Moran). The methodology used in this study could be applied to any city provided that census tracts are mapped and have certain demographic data recorded (all cities in the United States do), which could be useful for understanding of how local governments and other factors can play a role in the way pedestrian facilities are implemented.

Conclusion

The implementation of pedestrian safety measures such as crosswalks is clearly very important for reducing the likelihood of motor vehicle collisions with pedestrians and the severity of these collisions when they occur. Based on the data collected in the study, many places in Evansville lack sufficient crosswalk coverage, and they are far behind more dense cities such as San Francisco. Additionally, pedestrian facilities are even less prevalent in suburban and rural areas, making them more dangerous for pedestrians than the city center. The findings of this study indicate that by proactively installing pedestrian facilities such as crosswalks in areas that lack coverage, street designers can greatly improve upon existing infrastructure. This would aid in alleviating the growing number of pedestrian injuries and fatalities (IIHS). While this research provides insight into the crosswalk coverage of one city, it is important to consider the limitations in its scope and accuracy. By refining the methodology used and applying it in new locations, street designers could find areas which require the installation of more crosswalks.

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